

KISSsoft - Release 03/2017

Team-SolidSQUAD

Файл

Имя : GP
Изменил:: Admin дата: 01.03.2026 в: 18:54:24

АНАЛИЗ РЕЗУЛЬТАТОВ(Оценка важных свойств)

Комментарий:

№ = номер варианта
diff_i = отклонение от заданного передаточного отношения в %
кг = вес в кг
слайд = удельное скольжение (максимальное значение)
v.Slide = Скорость скольжения (м/с, максимальное значение)
AC/AE = Рабочий участок линии зацепления на входе AC / рабочий участок линии зацепления AE
(характеристики при трении)
del_cg = Стандартное отклонение жесткости при обкатке (Н/мм/мум)
1-eta = Потери в % (1.0-общий КПД)
Safety = запас прочности (ножка зуба и боковая поверхность, 0 = высоко, 1 = достаточно, 2 = низко)
(SF-min: 0.60/ 1.20/ 1.40 SH-min: 0.60/ 0.90/ 1.00)
Summary = Общая оценка (оценочно)
(50.0%:del_cg 20.0%:diff_i 100.0%:kg 35.0%:Slide 0.0%:v.Slide
0.0%:AC/AE 10.0%:1-eta 100.0%:Safety)

(Для таблицы справедливо: Чем меньше число, тем лучше!)

No.	diff_i	kg	Slide	v.Slide	AC/AE	del_cg	1-eta	Safety	Summary
1	-4.321	7.372	2.547	0.868	0.480	0.000	1.332	0.801	0.390
2	-4.321	7.343	1.876	0.932	0.434	0.000	1.348	0.810	0.390
3	-4.321	7.519	2.010	0.820	0.490	0.000	1.242	0.768	0.374
4	-4.321	7.490	1.549	0.883	0.444	0.000	1.252	0.777	0.375
5	-4.321	7.461	1.197	0.945	0.397	0.000	1.296	0.785	0.376
6	-2.778	7.542	2.775	0.881	0.476	0.000	1.360	0.808	0.390

7	-4.321	7.667	1.642	0.770	0.500	0.000	1.162	0.750	0.365
8	-4.321	7.638	1.304	0.832	0.454	0.000	1.167	0.758	0.366
9	-4.321	7.608	1.033	0.893	0.406	0.000	1.203	0.765	0.368
10	-2.778	7.691	2.164	0.836	0.486	0.000	1.268	0.774	0.373
11	-2.778	7.661	1.644	0.899	0.440	0.000	1.280	0.784	0.374
12	-2.778	7.631	1.255	0.961	0.393	0.000	1.327	0.793	0.375
13	-2.778	7.841	1.754	0.787	0.496	0.000	1.185	0.753	0.363
14	-2.778	7.811	1.379	0.849	0.450	0.000	1.192	0.761	0.364
15	-2.778	7.781	1.083	0.910	0.402	0.000	1.232	0.769	0.365
16	-1.235	7.865	2.337	0.850	0.482	0.000	1.295	0.781	0.373
17	-1.235	7.834	1.746	0.914	0.436	0.000	1.309	0.793	0.374
18	-1.235	7.804	1.316	0.975	0.389	0.000	1.359	0.801	0.375
19	-2.778	7.933	0.939	0.858	0.411	0.000	1.148	0.756	0.359
20	-1.235	8.017	1.878	0.803	0.492	0.000	1.210	0.756	0.360
21	-1.235	7.987	1.460	0.866	0.446	0.000	1.219	0.765	0.362
22	-1.235	7.956	1.136	0.927	0.399	0.000	1.261	0.773	0.363
23	0.309	8.041	2.531	0.864	0.478	0.000	1.321	0.791	0.375
24	0.309	8.010	1.856	0.927	0.432	0.000	1.338	0.803	0.376
25	-1.235	8.171	1.551	0.759	0.502	0.000	1.134	0.743	0.354
26	-1.235	8.140	1.238	0.815	0.455	0.000	1.139	0.750	0.355
27	-1.235	8.110	0.985	0.876	0.407	0.000	1.174	0.757	0.356
28	0.309	8.195	2.015	0.819	0.488	0.000	1.235	0.761	0.361
29	0.309	8.164	1.546	0.882	0.442	0.000	1.246	0.771	0.362
30	0.309	8.133	1.190	0.943	0.395	0.000	1.291	0.779	0.363
31	1.852	8.219	2.756	0.877	0.474	0.000	1.349	0.802	0.387
32	-4.971	8.059	2.531	0.885	0.481	0.000	1.251	0.736	0.369
33	0.309	8.351	1.654	0.770	0.498	0.000	1.157	0.744	0.353
34	0.309	8.320	1.308	0.832	0.451	0.000	1.163	0.752	0.354
35	0.309	8.288	1.032	0.893	0.404	0.000	1.201	0.759	0.355
36	1.852	8.375	2.164	0.834	0.484	0.000	1.261	0.767	0.370
37	1.852	8.343	1.637	0.896	0.438	0.000	1.274	0.777	0.371
38	1.852	8.312	1.245	0.958	0.390	0.000	1.322	0.786	0.372
39	-4.971	8.212	2.008	0.841	0.490	0.000	1.168	0.710	0.355
40	-4.971	8.182	1.548	0.906	0.444	0.000	1.177	0.718	0.356
41	-4.971	8.152	1.197	0.969	0.398	0.000	1.218	0.725	0.356
42	1.852	8.533	1.765	0.787	0.493	0.000	1.180	0.747	0.360
43	1.852	8.501	1.381	0.849	0.447	0.000	1.189	0.755	0.361

44	1.852	8.469	1.081	0.909	0.400	0.000	1.229	0.763	0.363
45	3.395	8.557	2.333	0.848	0.480	0.000	1.286	0.775	0.380
46	3.395	8.525	1.736	0.910	0.434	0.000	1.302	0.785	0.381
47	3.395	8.493	1.303	0.972	0.386	0.000	1.353	0.794	0.382
48	-4.971	8.366	1.647	0.794	0.500	0.000	1.093	0.694	0.347
49	-4.971	8.336	1.309	0.857	0.454	0.000	1.097	0.701	0.348
50	-4.971	8.305	1.038	0.920	0.406	0.000	1.131	0.707	0.349
51	-3.509	8.392	2.157	0.855	0.486	0.000	1.192	0.715	0.354
52	-3.509	8.361	1.640	0.920	0.441	0.000	1.203	0.724	0.355
53	-3.509	8.330	1.253	0.983	0.394	0.000	1.246	0.731	0.355
54	1.852	8.660	1.180	0.798	0.457	0.000	1.113	0.743	0.356
55	1.852	8.628	0.942	0.858	0.409	0.000	1.146	0.750	0.357
56	3.395	8.717	1.886	0.803	0.489	0.000	1.204	0.751	0.368
57	3.395	8.684	1.460	0.865	0.443	0.000	1.215	0.759	0.369
58	3.395	8.652	1.131	0.925	0.396	0.000	1.258	0.767	0.370
59	4.938	8.741	2.521	0.861	0.476	0.000	1.312	0.784	0.391
60	4.938	8.708	1.842	0.924	0.430	0.000	1.330	0.795	0.391
61	-4.971	8.491	1.120	0.806	0.463	0.000	1.027	0.692	0.344
62	-4.971	8.461	0.904	0.868	0.415	0.000	1.055	0.698	0.345
63	-3.509	8.548	1.757	0.810	0.496	0.000	1.116	0.697	0.345
64	-3.509	8.517	1.382	0.873	0.450	0.000	1.121	0.704	0.346
65	-3.509	8.486	1.087	0.936	0.403	0.000	1.158	0.710	0.346
66	-2.047	8.574	2.322	0.869	0.483	0.000	1.217	0.722	0.354
67	-2.047	8.542	1.737	0.933	0.437	0.000	1.229	0.731	0.354
68	3.395	8.879	1.566	0.755	0.499	0.000	1.131	0.737	0.361
69	3.395	8.846	1.245	0.816	0.453	0.000	1.136	0.744	0.362
70	3.395	8.813	0.986	0.876	0.405	0.000	1.172	0.751	0.364
71	4.938	8.903	2.019	0.818	0.485	0.000	1.229	0.755	0.377
72	4.938	8.870	1.543	0.880	0.439	0.000	1.241	0.764	0.378
73	4.938	8.837	1.183	0.940	0.392	0.000	1.287	0.773	0.379
74	-3.509	8.706	1.466	0.777	0.505	0.000	1.047	0.686	0.340
75	-3.509	8.675	1.181	0.824	0.459	0.000	1.048	0.692	0.340
76	-3.509	8.643	0.947	0.886	0.411	0.000	1.079	0.699	0.341
77	-2.047	8.732	1.876	0.825	0.492	0.000	1.138	0.700	0.343
78	-2.047	8.700	1.459	0.889	0.446	0.000	1.146	0.707	0.343
79	-2.047	8.669	1.136	0.951	0.399	0.000	1.185	0.714	0.344
80	-0.585	8.758	2.511	0.881	0.479	0.000	1.241	0.730	0.354

81	4.938	9.067	1.667	0.771	0.495	0.000	1.153	0.739	0.369
82	4.938	9.033	1.313	0.833	0.449	0.000	1.160	0.746	0.369
83	4.938	9.000	1.032	0.893	0.401	0.000	1.199	0.754	0.371
84	-2.047	8.892	1.558	0.781	0.501	0.000	1.067	0.687	0.337
85	-2.047	8.860	1.245	0.840	0.455	0.000	1.071	0.694	0.337
86	-2.047	8.828	0.991	0.902	0.408	0.000	1.104	0.700	0.338
87	-0.585	8.918	2.005	0.839	0.488	0.000	1.161	0.705	0.342
88	-0.585	8.885	1.540	0.903	0.442	0.000	1.170	0.712	0.342
89	-0.585	8.853	1.187	0.965	0.395	0.000	1.212	0.719	0.343
90	4.938	9.165	0.903	0.842	0.410	0.000	1.121	0.744	0.367
91	-2.047	8.989	0.866	0.851	0.416	0.000	1.031	0.693	0.335
92	-0.585	9.080	1.654	0.793	0.497	0.000	1.088	0.690	0.334
93	-0.585	9.047	1.310	0.856	0.451	0.000	1.093	0.696	0.335
94	-0.585	9.014	1.035	0.918	0.404	0.000	1.129	0.702	0.336
95	0.877	9.106	2.149	0.852	0.484	0.000	1.184	0.710	0.346
96	0.877	9.073	1.628	0.916	0.439	0.000	1.195	0.718	0.347
97	0.877	9.040	1.240	0.979	0.391	0.000	1.240	0.725	0.347
98	-0.585	9.210	1.127	0.807	0.460	0.000	1.024	0.687	0.330
99	-0.585	9.177	0.907	0.869	0.412	0.000	1.054	0.693	0.332
100	0.877	9.269	1.762	0.809	0.493	0.000	1.110	0.692	0.337
101	0.877	9.236	1.381	0.872	0.447	0.000	1.117	0.699	0.338
102	0.877	9.203	1.082	0.934	0.400	0.000	1.154	0.706	0.339
103	2.339	9.295	2.309	0.865	0.481	0.000	1.208	0.716	0.356
104	2.339	9.262	1.722	0.929	0.435	0.000	1.221	0.725	0.356
105	-4.167	9.124	2.146	0.872	0.487	0.000	1.124	0.668	0.340
106	-4.167	9.092	1.633	0.938	0.441	0.000	1.133	0.675	0.340
107	0.877	9.435	1.477	0.770	0.503	0.000	1.043	0.681	0.332
108	0.877	9.401	1.185	0.824	0.456	0.000	1.046	0.688	0.333
109	0.877	9.368	0.948	0.885	0.409	0.000	1.077	0.694	0.334
110	2.339	9.461	1.877	0.823	0.490	0.000	1.132	0.695	0.345
111	2.339	9.427	1.455	0.886	0.444	0.000	1.141	0.703	0.346
112	2.339	9.394	1.129	0.948	0.397	0.000	1.181	0.710	0.346
113	-4.167	9.286	1.755	0.830	0.496	0.000	1.053	0.651	0.332
114	-4.167	9.254	1.382	0.895	0.450	0.000	1.058	0.657	0.332
115	-4.167	9.222	1.087	0.959	0.403	0.000	1.092	0.663	0.332
116	-2.778	9.314	2.305	0.884	0.484	0.000	1.146	0.674	0.340
117	2.339	9.628	1.566	0.778	0.499	0.000	1.063	0.683	0.339

118	2.339	9.594	1.247	0.840	0.453	0.000	1.068	0.689	0.340
119	2.339	9.561	0.990	0.901	0.405	0.000	1.101	0.696	0.341
120	3.801	9.655	2.005	0.837	0.486	0.000	1.155	0.700	0.353
121	3.801	9.621	1.535	0.900	0.440	0.000	1.165	0.707	0.354
122	3.801	9.587	1.179	0.962	0.393	0.000	1.208	0.714	0.354
123	-4.167	9.450	1.472	0.798	0.505	0.000	0.989	0.638	0.325
124	-4.167	9.417	1.186	0.848	0.458	0.000	0.990	0.645	0.326
125	-4.167	9.385	0.952	0.911	0.411	0.000	1.018	0.653	0.328
126	-2.778	9.478	1.871	0.844	0.492	0.000	1.075	0.654	0.330
127	-2.778	9.445	1.457	0.909	0.447	0.000	1.081	0.661	0.330
128	-2.778	9.412	1.135	0.973	0.400	0.000	1.117	0.667	0.330
129	2.339	9.729	0.870	0.852	0.413	0.000	1.031	0.688	0.338
130	3.801	9.824	1.662	0.793	0.495	0.000	1.084	0.685	0.346
131	3.801	9.790	1.312	0.856	0.449	0.000	1.090	0.691	0.346
132	3.801	9.755	1.033	0.917	0.402	0.000	1.126	0.698	0.347
133	-4.167	9.549	0.837	0.861	0.419	0.000	0.953	0.645	0.325
134	-2.778	9.644	1.560	0.802	0.501	0.000	1.008	0.640	0.323
135	-2.778	9.610	1.247	0.864	0.455	0.000	1.011	0.647	0.324
136	-2.778	9.577	0.994	0.927	0.408	0.000	1.041	0.654	0.325
137	-1.389	9.672	1.995	0.857	0.489	0.000	1.095	0.658	0.328
138	-1.389	9.638	1.535	0.922	0.443	0.000	1.103	0.665	0.328
139	-1.389	9.605	1.184	0.986	0.396	0.000	1.142	0.671	0.329
140	3.801	9.960	1.133	0.808	0.458	0.000	1.022	0.683	0.343
141	3.801	9.925	0.909	0.869	0.410	0.000	1.053	0.689	0.344
142	-2.778	9.777	1.079	0.815	0.463	0.000	0.947	0.639	0.320
143	-2.778	9.744	0.874	0.878	0.416	0.000	0.973	0.645	0.321
144	-1.389	9.839	1.654	0.814	0.497	0.000	1.028	0.642	0.320
145	-1.389	9.805	1.311	0.878	0.451	0.000	1.032	0.650	0.322
146	-1.389	9.772	1.037	0.942	0.404	0.000	1.065	0.656	0.322
147	0.000	9.867	2.133	0.869	0.486	0.000	1.116	0.663	0.327
148	0.000	9.833	1.618	0.934	0.440	0.000	1.125	0.671	0.327
149	-1.389	10.008	1.399	0.787	0.506	0.000	0.966	0.633	0.316
150	-1.389	9.974	1.133	0.832	0.460	0.000	0.967	0.639	0.317
151	-1.389	9.940	0.912	0.894	0.412	0.000	0.994	0.646	0.318
152	0.000	10.037	1.757	0.828	0.494	0.000	1.048	0.645	0.318
153	0.000	10.002	1.379	0.893	0.448	0.000	1.054	0.653	0.319
154	0.000	9.968	1.081	0.956	0.401	0.000	1.088	0.659	0.320

155	-4.762	9.888	2.132	0.887	0.488	0.000	1.063	0.624	0.327
156	0.000	10.208	1.480	0.792	0.502	0.000	0.985	0.634	0.313
157	0.000	10.173	1.190	0.848	0.456	0.000	0.987	0.640	0.314
158	0.000	10.139	0.952	0.910	0.409	0.000	1.017	0.647	0.315
159	1.389	10.236	1.868	0.841	0.490	0.000	1.068	0.649	0.326
160	1.389	10.201	1.450	0.906	0.445	0.000	1.075	0.657	0.327
161	1.389	10.167	1.127	0.969	0.397	0.000	1.112	0.663	0.327
162	-4.762	10.056	1.753	0.848	0.496	0.000	0.998	0.577	0.306
163	-4.762	10.022	1.381	0.914	0.450	0.000	1.002	0.586	0.308
164	-4.762	9.988	1.087	0.980	0.403	0.000	1.034	0.599	0.311
165	0.000	10.311	0.839	0.862	0.417	0.000	0.952	0.641	0.313
166	1.389	10.409	1.565	0.799	0.499	0.000	1.004	0.635	0.320
167	1.389	10.374	1.248	0.863	0.453	0.000	1.007	0.642	0.320
168	1.389	10.339	0.992	0.925	0.406	0.000	1.039	0.650	0.322
169	2.778	10.438	1.989	0.854	0.487	0.000	1.088	0.654	0.334
170	2.778	10.403	1.525	0.918	0.441	0.000	1.097	0.661	0.334
171	-4.762	10.225	1.475	0.819	0.504	0.000	0.937	0.566	0.301
172	-4.762	10.191	1.190	0.870	0.458	0.000	0.938	0.573	0.302
173	-4.762	10.157	0.956	0.935	0.411	0.000	0.965	0.581	0.304
174	-3.439	10.255	1.863	0.860	0.493	0.000	1.017	0.582	0.305
175	-3.439	10.221	1.452	0.927	0.447	0.000	1.022	0.593	0.307
176	-3.439	10.187	1.132	0.992	0.400	0.000	1.056	0.613	0.314
177	1.389	10.549	1.084	0.816	0.461	0.000	0.945	0.634	0.317
178	1.389	10.513	0.876	0.878	0.413	0.000	0.972	0.641	0.318
179	2.778	10.613	1.658	0.813	0.495	0.000	1.023	0.637	0.326
180	2.778	10.577	1.310	0.877	0.449	0.000	1.028	0.644	0.327
181	2.778	10.542	1.033	0.940	0.402	0.000	1.062	0.652	0.329
182	4.167	10.642	2.122	0.866	0.484	0.000	1.109	0.659	0.343
183	-4.762	10.362	1.035	0.823	0.466	0.000	0.880	0.568	0.300
184	-4.762	10.328	0.844	0.887	0.419	0.000	0.902	0.575	0.301
185	-3.439	10.426	1.561	0.822	0.501	0.000	0.956	0.567	0.298
186	-3.439	10.392	1.249	0.885	0.455	0.000	0.957	0.574	0.299
187	-3.439	10.358	0.996	0.949	0.408	0.000	0.986	0.582	0.301
188	-2.116	10.457	1.983	0.872	0.490	0.000	1.036	0.588	0.305
189	2.778	10.790	1.408	0.782	0.504	0.000	0.963	0.629	0.322
190	2.778	10.754	1.137	0.832	0.458	0.000	0.965	0.635	0.323
191	2.778	10.718	0.913	0.894	0.410	0.000	0.993	0.641	0.324

192	4.167	10.819	1.758	0.827	0.492	0.000	1.043	0.640	0.334
193	4.167	10.782	1.375	0.891	0.446	0.000	1.049	0.648	0.334
194	4.167	10.746	1.075	0.953	0.399	0.000	1.085	0.655	0.336
195	-3.439	10.599	1.330	0.804	0.509	0.000	0.899	0.562	0.295
196	-3.439	10.565	1.085	0.840	0.463	0.000	0.898	0.568	0.296
197	-3.439	10.530	0.880	0.903	0.416	0.000	0.922	0.574	0.298
198	-2.116	10.630	1.652	0.833	0.498	0.000	0.974	0.569	0.296
199	-2.116	10.595	1.310	0.898	0.452	0.000	0.977	0.576	0.297
200	-2.116	10.560	1.037	0.963	0.405	0.000	1.007	0.586	0.299
201	4.167	10.997	1.487	0.785	0.500	0.000	0.981	0.630	0.328
202	4.167	10.961	1.191	0.847	0.454	0.000	0.984	0.636	0.329
203	4.167	10.924	0.951	0.909	0.407	0.000	1.015	0.643	0.330
204	-2.116	10.805	1.403	0.808	0.506	0.000	0.916	0.562	0.292
205	-2.116	10.769	1.136	0.855	0.460	0.000	0.916	0.568	0.293
206	-2.116	10.734	0.916	0.918	0.412	0.000	0.942	0.575	0.295
207	-0.794	10.835	1.750	0.846	0.494	0.000	0.992	0.572	0.294
208	-0.794	10.800	1.375	0.912	0.448	0.000	0.997	0.580	0.295
209	-0.794	10.764	1.079	0.976	0.401	0.000	1.029	0.590	0.298
210	4.167	11.104	0.842	0.863	0.414	0.000	0.951	0.637	0.328
211	-2.116	10.910	0.811	0.871	0.420	0.000	0.882	0.571	0.293
212	-0.794	11.012	1.480	0.812	0.502	0.000	0.933	0.562	0.290
213	-0.794	10.976	1.190	0.869	0.456	0.000	0.935	0.568	0.290
214	-0.794	10.940	0.953	0.933	0.409	0.000	0.962	0.576	0.292
215	0.529	11.043	1.857	0.858	0.491	0.000	1.010	0.575	0.296
216	0.529	11.007	1.443	0.923	0.445	0.000	1.016	0.585	0.298
217	-0.794	11.191	1.270	0.793	0.511	0.000	0.879	0.559	0.288
218	-0.794	11.154	1.039	0.824	0.464	0.000	0.878	0.564	0.289
219	-0.794	11.118	0.845	0.887	0.417	0.000	0.901	0.571	0.290
220	0.529	11.221	1.563	0.818	0.499	0.000	0.951	0.563	0.290
221	0.529	11.185	1.247	0.883	0.453	0.000	0.954	0.570	0.291
222	0.529	11.149	0.992	0.947	0.406	0.000	0.983	0.577	0.293
223	1.852	11.252	1.972	0.869	0.488	0.000	1.029	0.581	0.305
224	-4.040	11.063	1.853	0.875	0.494	0.000	0.965	0.541	0.294
225	0.529	11.402	1.336	0.797	0.507	0.000	0.896	0.558	0.288
226	0.529	11.365	1.087	0.839	0.461	0.000	0.895	0.564	0.289
227	0.529	11.328	0.879	0.902	0.413	0.000	0.920	0.571	0.290
228	1.852	11.433	1.652	0.831	0.496	0.000	0.969	0.565	0.297

229	1.852	11.396	1.307	0.896	0.450	0.000	0.973	0.572	0.298
230	1.852	11.359	1.032	0.960	0.403	0.000	1.004	0.580	0.299
231	-4.040	11.240	1.560	0.841	0.501	0.000	0.908	0.533	0.290
232	-4.040	11.204	1.249	0.904	0.455	0.000	0.909	0.538	0.290
233	-4.040	11.169	0.997	0.970	0.408	0.000	0.936	0.543	0.290
234	1.852	11.615	1.408	0.802	0.504	0.000	0.913	0.558	0.294
235	1.852	11.578	1.138	0.854	0.458	0.000	0.913	0.564	0.294
236	1.852	11.541	0.915	0.917	0.410	0.000	0.940	0.571	0.296
237	3.175	11.646	1.746	0.843	0.493	0.000	0.987	0.567	0.304
238	3.175	11.609	1.368	0.909	0.447	0.000	0.991	0.575	0.304
239	3.175	11.571	1.071	0.972	0.400	0.000	1.025	0.584	0.306
240	-4.040	11.419	1.334	0.824	0.509	0.000	0.855	0.531	0.288
241	-4.040	11.383	1.089	0.861	0.463	0.000	0.853	0.535	0.289
242	-4.040	11.347	0.883	0.926	0.416	0.000	0.876	0.540	0.289
243	-2.778	11.451	1.646	0.849	0.498	0.000	0.925	0.534	0.287
244	-2.778	11.415	1.307	0.916	0.452	0.000	0.927	0.538	0.287
245	-2.778	11.379	1.035	0.982	0.405	0.000	0.955	0.544	0.287
246	1.852	11.761	0.997	0.809	0.465	0.000	0.859	0.562	0.293
247	1.852	11.724	0.813	0.871	0.417	0.000	0.882	0.568	0.294
248	3.175	11.831	1.485	0.805	0.500	0.000	0.930	0.558	0.299
249	3.175	11.793	1.191	0.868	0.454	0.000	0.932	0.565	0.300
250	3.175	11.755	0.952	0.931	0.407	0.000	0.960	0.571	0.301
251	4.497	11.862	1.851	0.855	0.490	0.000	1.005	0.570	0.311
252	4.497	11.824	1.435	0.920	0.444	0.000	1.011	0.579	0.312
253	-4.040	11.563	0.955	0.816	0.470	0.000	0.802	0.535	0.288
254	-4.040	11.527	0.785	0.880	0.422	0.000	0.821	0.540	0.289
255	-2.778	11.632	1.404	0.828	0.506	0.000	0.871	0.530	0.285
256	-2.778	11.595	1.139	0.876	0.460	0.000	0.871	0.535	0.285
257	-2.778	11.559	0.919	0.940	0.413	0.000	0.895	0.539	0.286
258	-1.515	11.665	1.742	0.861	0.495	0.000	0.941	0.535	0.285
259	-1.515	11.628	1.369	0.928	0.449	0.000	0.945	0.540	0.285
260	3.175	12.017	1.278	0.788	0.509	0.000	0.877	0.555	0.298
261	3.175	11.978	1.043	0.824	0.462	0.000	0.876	0.561	0.299
262	3.175	11.940	0.846	0.887	0.414	0.000	0.900	0.567	0.300
263	4.497	12.048	1.565	0.817	0.497	0.000	0.947	0.559	0.305
264	4.497	12.009	1.245	0.881	0.451	0.000	0.950	0.566	0.306
265	4.497	11.971	0.988	0.945	0.404	0.000	0.980	0.573	0.307

266	-2.778	11.814	1.212	0.808	0.514	0.000	0.821	0.530	0.285
267	-2.778	11.777	0.998	0.831	0.467	0.000	0.818	0.534	0.285
268	-2.778	11.741	0.817	0.895	0.419	0.000	0.839	0.539	0.286
269	-1.515	11.847	1.478	0.831	0.502	0.000	0.887	0.530	0.282
270	-1.515	11.810	1.190	0.889	0.457	0.000	0.888	0.535	0.282
271	-1.515	11.773	0.954	0.954	0.410	0.000	0.913	0.539	0.283
272	-0.253	11.880	1.843	0.872	0.492	0.000	0.958	0.537	0.283
273	4.497	12.236	1.344	0.792	0.505	0.000	0.893	0.555	0.303
274	4.497	12.197	1.091	0.839	0.459	0.000	0.894	0.561	0.304
275	4.497	12.159	0.880	0.902	0.411	0.000	0.919	0.567	0.305
276	-1.515	12.032	1.274	0.813	0.510	0.000	0.836	0.529	0.282
277	-1.515	11.994	1.043	0.846	0.464	0.000	0.834	0.534	0.282
278	-1.515	11.957	0.849	0.910	0.417	0.000	0.856	0.538	0.282
279	-0.253	12.064	1.558	0.835	0.499	0.000	0.903	0.531	0.280
280	-0.253	12.027	1.244	0.901	0.454	0.000	0.905	0.535	0.280
281	-0.253	11.989	0.990	0.966	0.407	0.000	0.932	0.540	0.280
282	4.497	12.347	0.784	0.856	0.418	0.000	0.863	0.565	0.305
283	-1.515	12.142	0.756	0.864	0.423	0.000	0.804	0.538	0.283
284	-0.253	12.251	1.339	0.817	0.507	0.000	0.852	0.528	0.278
285	-0.253	12.213	1.090	0.861	0.461	0.000	0.851	0.533	0.279
286	-0.253	12.175	0.882	0.925	0.414	0.000	0.874	0.537	0.279
287	1.010	12.284	1.644	0.847	0.496	0.000	0.920	0.531	0.284
288	1.010	12.245	1.302	0.914	0.451	0.000	0.922	0.535	0.284
289	1.010	12.207	1.028	0.979	0.404	0.000	0.951	0.541	0.284
290	-4.589	12.085	1.556	0.858	0.502	0.000	0.865	0.509	0.285
291	-4.589	12.048	1.247	0.921	0.456	0.000	0.865	0.512	0.284
292	-4.589	12.011	0.995	0.988	0.409	0.000	0.890	0.516	0.285
293	-0.253	12.400	0.960	0.816	0.468	0.000	0.801	0.533	0.279
294	-0.253	12.362	0.787	0.880	0.420	0.000	0.821	0.537	0.279
295	1.010	12.472	1.407	0.821	0.504	0.000	0.867	0.528	0.282
296	1.010	12.433	1.138	0.874	0.458	0.000	0.868	0.532	0.282
297	1.010	12.395	0.916	0.939	0.411	0.000	0.893	0.537	0.283
298	2.273	12.505	1.735	0.858	0.494	0.000	0.936	0.533	0.290
299	2.273	12.466	1.360	0.925	0.448	0.000	0.939	0.537	0.289
300	-4.589	12.270	1.337	0.843	0.509	0.000	0.815	0.507	0.284
301	-4.589	12.232	1.092	0.881	0.463	0.000	0.813	0.511	0.284
302	-4.589	12.195	0.886	0.947	0.416	0.000	0.835	0.515	0.284

303	-3.382	12.304	1.640	0.864	0.499	0.000	0.880	0.509	0.283
304	-3.382	12.267	1.302	0.932	0.453	0.000	0.881	0.512	0.282
305	1.010	12.661	1.218	0.803	0.512	0.000	0.818	0.528	0.282
306	1.010	12.623	1.001	0.831	0.465	0.000	0.816	0.532	0.282
307	1.010	12.584	0.817	0.895	0.418	0.000	0.837	0.537	0.283
308	2.273	12.695	1.480	0.824	0.501	0.000	0.883	0.528	0.287
309	2.273	12.656	1.189	0.887	0.455	0.000	0.884	0.532	0.287
310	2.273	12.617	0.951	0.952	0.408	0.000	0.911	0.537	0.288
311	3.535	12.728	1.834	0.869	0.491	0.000	0.952	0.534	0.296
312	-4.589	12.456	1.160	0.823	0.516	0.000	0.769	0.509	0.284
313	-4.589	12.418	0.961	0.838	0.470	0.000	0.765	0.512	0.285
314	-4.589	12.381	0.790	0.903	0.422	0.000	0.783	0.516	0.285
315	-3.382	12.491	1.404	0.846	0.506	0.000	0.830	0.507	0.281
316	-3.382	12.453	1.139	0.894	0.460	0.000	0.829	0.510	0.281
317	-3.382	12.415	0.920	0.960	0.413	0.000	0.852	0.514	0.281
318	-2.174	12.525	1.731	0.875	0.496	0.000	0.895	0.510	0.281

АНАЛИЗ РЕЗУЛЬТАТОВ

(с указанием номеров вариантов в нисходящей последовательности)

Лучшие варианты относительно передаточного отношения: 147 148 152 153 154 156 157 158 165 272 ...

Лучшие варианты относительно наименьшего веса: 2 1 5 4 3 6 9 12 8 11 ...

Оптимальные варианты относительно трения (AC/AE): 47 18 38 97 73 12 122 53 30 89 ...

Лучшие варианты относительно вариации жесткости: 1 2 4 6 8 11 12 15 21 26 ...

Лучшие варианты относительно прочности: 315 300 312 290 303 318 316 301 291 304 ...

Лучшие варианты в общем (сумма) : 284 285 293 286 294 280 279 281 318 316 ...

ОПРЕДЕЛЕНИЕ ВАРИАНТОВ ЗУБЧАТОГО ЗАЦЕПЛЕНИЯ

Вводные данные!

Заданное передаточное отношение 3.600 +/- 5.000 % (3.420 ... 3.780)

Мощность: 23.9599 kW

Исходная частота вращения: 1040.00 (Колесо 1)

Модуль 2.5000 mm
 Угол зацепления 20.000
 Угол наклона линии зуба 23.000
 Межосевое расстояние от 100.0000 mm до 140.0000 mm с шагом 1.0000 mm
 Смещение исходного контура допустимо в диапазоне -0.500 ... 0.500
 Минимальное число зубьев 18
 >> Отбрасываются результаты с удельным скольжением выше (zeta > 3.000; zeta < -3.000).
 >> Результаты толщины зуба на головке < минимальная толщина зуба отбрасываются.
 x1(max)-criteria: согласно скомпенсированному удельному скольжению

>> Момент инерции: Выдаются данные для области делительной окружности до внутреннего диаметра (при внутреннем зацеплении для делительной окружности до наружного диаметра)
 >> Варианты, не отвечающие заданным запасам прочности, отклоняются.

Комментарий:

№ = номер варианта
 a = Межосевое расстояние
 mn = нормальный модуль
 alfa = Угол зацепления
 бета = угол наклона линии зуба
 Z1, Z2 = число зубьев колес 1, 2
 X1+X2, X1, X2 = смещение исходного контура, сумма, колесо 1, колесо 2
 Ea = Торцевое перекрытие (eps_a.e+eps_a.i)/2
 Eb = Осевое перекрытие
 Eg = общее перекрытие (eps_g.e+eps_g.i)/2
 * = Варианты, отмеченные *, являются оптимальными (целочисленные общие перекрытия)
 слайд = удельное скольжение (максимальное значение)
 i = передаточное отношение
 diff_i = отклонение от заданного передаточного отношения в %
 SF1, SF2 = запас прочности ножки зуба колес 1, 2
 SF1, SF2 = запас прочности боковой поверхности (в однопарном зацеплении) колес 1, 2
 SB = запас прочности на заедание, температура «вспышки» в пятне контакта
 SInt = запас прочности на заедание, интегральная температура
 del_cg = Стандартное отклонение жесткости при обкатке (Н/мм/мум)

No.	a	mn	alfa	beta	Z1	Z2	X1+X2	X1	Ea	Eb	Eg	Slide	i	diff_i	SF1
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SF2	SH1	SH2SB	SInt	del_cg																			
1	108.00	2.50	20.0	23.0	18	62	-0.250	+0.070	1.49	1.99	3.48*-2.55	-1.17	3.44	-4.3		2.65	2.72	1.08	1.12	3.80	3.26	0.00	
2								+0.170	1.47	1.99	3.46*-1.88	-1.28				2.64	2.60	1.08	1.12	4.30	3.26	0.00	
3	109.00	2.50	20.0	23.0	18	62	+0.147	+0.128	1.44	1.99	3.43*-2.01	-1.00	3.44	-4.3		2.52	2.56	1.10	1.14	4.13	3.34	0.00	
4								+0.228	1.42	1.99	3.41*-1.55	-1.10				2.56	2.52	1.09	1.13	4.64	3.34	0.00	
5								+0.328	1.40	1.99	3.39*-1.20	-1.20				2.59	2.48	1.09	1.13	4.47	3.31	0.00	
6	109.00	2.50	20.0	23.0	18	63	-0.386	+0.055	1.51	1.99	3.50*-2.78	-1.23	3.50	-2.8		2.62	2.68	1.08	1.12	3.71	3.24	0.00	
7	110.00	2.50	20.0	23.0	18	62	+0.567	+0.197	1.38	1.99	3.37*-1.64	-0.86	3.44	-4.3		2.47	2.48	1.11	1.15	4.45	3.41	0.00	
8								+0.297	1.36	1.99	3.35*-1.30	-0.95				2.50	2.45	1.10	1.15	4.97	3.41	0.00	
9								+0.397	1.35	1.99	3.34*-1.03	-1.03				2.53	2.42	1.10	1.14	4.85	3.39	0.00	
10	110.00	2.50	20.0	23.0	18	63	+0.002	+0.108	1.46	1.99	3.45*-2.16	-1.05	3.50	-2.8		2.54	2.59	1.10	1.14	4.04	3.33	0.00	
11								+0.208	1.44	1.99	3.43*-1.64	-1.15				2.59	2.55	1.09	1.13	4.54	3.32	0.00	
12								+0.308	1.42	1.99	3.41*-1.26	-1.25				2.62	2.51	1.08	1.12	4.35	3.29	0.00	
13	111.00	2.50	20.0	23.0	18	63	+0.414	+0.172	1.40	1.99	3.39*-1.75	-0.91	3.50	-2.8		2.48	2.50	1.11	1.15	4.36	3.39	0.00	
14								+0.272	1.38	1.99	3.37*-1.38	-0.99				2.52	2.47	1.10	1.15	4.88	3.40	0.00	
15								+0.372	1.37	1.99	3.36*-1.08	-1.08				2.55	2.44	1.10	1.14	4.73	3.37	0.00	
16	111.00	2.50	20.0	23.0	18	64	-0.139	+0.089	1.48	1.99	3.47*-2.34	-1.11	3.56	-1.2		2.57	2.62	1.09	1.13	3.94	3.31	0.00	
17								+0.189	1.46	1.99	3.45*-1.75	-1.21				2.62	2.58	1.08	1.13	4.45	3.30	0.00	
18								+0.289	1.44	1.99	3.43*-1.32	-1.32				2.65	2.53	1.08	1.12	4.24	3.26	0.00	
19	112.00	2.50	20.0	23.0	18	63	+0.847	+0.447	1.31	1.99	3.30*-0.94	-0.94	3.50	-2.8		2.50	2.38	1.10	1.15	5.12	3.43	0.00	
20	112.00	2.50	20.0	23.0	18	64	+0.264	+0.149	1.42	1.99	3.41*-1.88	-0.95	3.56	-1.2		2.50	2.53	1.11	1.15	4.26	3.38	0.00	
21								+0.249	1.41	1.99	3.40*-1.46	-1.04				2.54	2.50	1.10	1.14	4.78	3.38	0.00	
22								+0.349	1.39	1.99	3.38*-1.14	-1.13				2.57	2.46	1.09	1.14	4.61	3.35	0.00	
23	112.00	2.50	20.0	23.0	18	65	-0.278	+0.073	1.50	1.99	3.49*-2.53	-1.16	3.61	+0.3		2.59	2.65	1.09	1.13	3.85	3.29	0.00	
24								+0.173	1.47	1.99	3.46*-1.86	-1.27				2.65	2.60	1.08	1.12	4.35	3.28	0.00	
25	113.00	2.50	20.0	23.0	18	64	+0.689	+0.219	1.36	1.99	3.35*-1.55	-0.82	3.56	-1.2		2.46	2.46	1.12	1.16	4.58	3.44	0.00	
26								+0.319	1.35	1.99	3.34*-1.24	-0.90				2.49	2.43	1.11	1.15	5.11	3.44	0.00	
27								+0.419	1.33	1.99	3.32*-0.98	-0.98				2.51	2.40	1.10	1.15	4.99	3.42	0.00	
28	113.00	2.50	20.0	23.0	18	65	+0.117	+0.128	1.44	1.99	3.43*-2.02	-1.00	3.61	+0.3		2.53	2.56	1.10	1.15	4.17	3.36	0.00	
29								+0.228	1.43	1.99	3.42*-1.55	-1.09				2.57	2.52	1.10	1.14	4.68	3.36	0.00	
30								+0.328	1.41	1.99	3.40*-1.19	-1.19				2.60	2.49	1.09	1.13	4.49	3.33	0.00	

31	113.00	2.50	20.0	23.0	18	66	-0.414	+0.058	1.51	1.99	3.50*	-2.76	-1.22	3.67	+1.9	2.62	2.69	1.08	1.12	3.75	3.26	0.00
32	113.00	2.50	20.0	23.0	19	65	-0.414	+0.032	1.52	1.99	3.51*	-2.53	-1.17	3.42	-5.0	2.72	2.79	1.12	1.17	4.01	3.35	0.00
33	114.00	2.50	20.0	23.0	18	65	+0.534	+0.193	1.38	1.99	3.37*	-1.65	-0.86	3.61	+0.3	2.47	2.48	1.11	1.16	4.48	3.43	0.00
34																2.51	2.45	1.11	1.15	5.01	3.43	0.00
35																2.53	2.42	1.10	1.15	4.87	3.40	0.00
36	114.00	2.50	20.0	23.0	18	66	-0.027	+0.109	1.46	1.99	3.45*	-2.16	-1.05	3.67	+1.9	2.55	2.59	1.10	1.14	4.07	3.35	0.00
37																2.60	2.55	1.09	1.14	4.59	3.34	0.00
38																2.63	2.51	1.09	1.13	4.38	3.30	0.00
39	114.00	2.50	20.0	23.0	19	65	-0.027	+0.089	1.47	1.99	3.46*	-2.01	-1.00	3.42	-5.0	2.64	2.69	1.14	1.19	4.35	3.43	0.00
40																2.69	2.65	1.14	1.18	4.88	3.43	0.00
41																2.73	2.61	1.13	1.17	4.70	3.39	0.00
42	115.00	2.50	20.0	23.0	18	66	+0.382	+0.169	1.41	1.99	3.40*	-1.76	-0.91	3.67	+1.9	2.49	2.51	1.11	1.16	4.39	3.41	0.00
43																2.53	2.47	1.11	1.15	4.91	3.41	0.00
44																2.56	2.44	1.10	1.14	4.75	3.38	0.00
45	115.00	2.50	20.0	23.0	18	67	-0.168	+0.091	1.48	1.99	3.47*	-2.33	-1.10	3.72	+3.4	2.57	2.62	1.10	1.14	3.98	3.33	0.00
46																2.62	2.57	1.09	1.13	4.50	3.32	0.00
47																2.66	2.53	1.08	1.13	4.28	3.28	0.00
48	115.00	2.50	20.0	23.0	19	65	+0.382	+0.155	1.42	1.99	3.41*	-1.65	-0.87	3.42	-5.0	2.58	2.60	1.16	1.20	4.69	3.49	0.00
49																2.62	2.57	1.15	1.19	5.23	3.50	0.00
50																2.65	2.53	1.14	1.19	5.09	3.47	0.00
51	115.00	2.50	20.0	23.0	19	66	-0.168	+0.069	1.49	1.99	3.48*	-2.16	-1.05	3.47	-3.5	2.67	2.73	1.14	1.18	4.25	3.41	0.00
52																2.72	2.68	1.13	1.18	4.78	3.41	0.00
53																2.76	2.63	1.13	1.17	4.58	3.37	0.00
54	116.00	2.50	20.0	23.0	18	66	+0.813	+0.340	1.33	1.99	3.32*	-1.18	-0.86	3.67	+1.9	2.48	2.41	1.11	1.16	5.24	3.47	0.00
55																2.50	2.38	1.11	1.15	5.13	3.45	0.00
56	116.00	2.50	20.0	23.0	18	67	+0.233	+0.147	1.43	1.99	3.42*	-1.89	-0.95	3.72	+3.4	2.51	2.53	1.11	1.16	4.29	3.40	0.00
57																2.55	2.49	1.10	1.15	4.82	3.40	0.00
58																2.58	2.46	1.10	1.14	4.63	3.36	0.00
59	116.00	2.50	20.0	23.0	18	68	-0.307	+0.076	1.50	1.99	3.49*	-2.52	-1.15	3.78	+4.9	2.60	2.65	1.09	1.14	3.89	3.31	0.00
60																2.66	2.60	1.08	1.13	4.41	3.30	0.00
61	116.00	2.50	20.0	23.0	19	65	+0.813	+0.332	1.34	1.99	3.33*	-1.12	-0.83	3.42	-5.0	2.57	2.50	1.16	1.20	5.58	3.55	0.00
62																2.59	2.47	1.15	1.20	5.50	3.53	0.00
63	116.00	2.50	20.0	23.0	19	66	+0.233	+0.131	1.44	1.99	3.43*	-1.76	-0.91	3.47	-3.5	2.60	2.63	1.15	1.20	4.59	3.48	0.00
64																2.64	2.59	1.15	1.19	5.13	3.48	0.00

65								+0.331	1.40	1.99	3.39*-1.09-1.09		2.67	2.56	1.14	1.19	4.96	3.45	0.00
66	116.00	2.50	20.0	23.0	19	67	-0.307	+0.052	1.51	1.99	3.50*-2.32-1.10	3.53-2.0	2.70	2.75	1.14	1.18	4.16	3.39	0.00
67								+0.152	1.49	1.99	3.48*-1.74-1.20		2.75	2.70	1.13	1.17	4.69	3.38	0.00
68	117.00	2.50	20.0	23.0	18	67	+0.656	+0.214	1.37	1.99	3.36*-1.57-0.83	3.72+3.4	2.47	2.45	1.12	1.17	4.60	3.45	0.00
69								+0.314	1.35	1.99	3.34*-1.24-0.91		2.50	2.42	1.11	1.16	5.14	3.46	0.00
70								+0.414	1.34	1.99	3.33*-0.99-0.99		2.52	2.39	1.11	1.15	5.00	3.43	0.00
71	117.00	2.50	20.0	23.0	18	68	+0.087	+0.127	1.45	1.99	3.44*-2.02-1.00	3.78+4.9	2.53	2.56	1.11	1.15	4.20	3.38	0.00
72								+0.227	1.43	1.99	3.42*-1.54-1.09		2.58	2.52	1.10	1.15	4.73	3.38	0.00
73								+0.327	1.41	1.99	3.40*-1.18-1.18		2.61	2.48	1.09	1.14	4.52	3.34	0.00
74	117.00	2.50	20.0	23.0	19	66	+0.656	+0.204	1.38	1.99	3.37*-1.47-0.79	3.47-3.5	2.56	2.55	1.16	1.21	4.92	3.54	0.00
75								+0.304	1.36	1.99	3.35*-1.18-0.87		2.59	2.52	1.16	1.20	5.48	3.54	0.00
76								+0.404	1.35	1.99	3.34*-0.95-0.95		2.61	2.49	1.15	1.20	5.36	3.52	0.00
77	117.00	2.50	20.0	23.0	19	67	+0.087	+0.109	1.46	1.99	3.45*-1.88-0.95	3.53-2.0	2.63	2.65	1.15	1.20	4.49	3.46	0.00
78								+0.209	1.44	1.99	3.43*-1.46-1.04		2.67	2.62	1.15	1.19	5.03	3.46	0.00
79								+0.309	1.42	1.99	3.41*-1.14-1.14		2.70	2.58	1.14	1.18	4.85	3.43	0.00
80	117.00	2.50	20.0	23.0	19	68	-0.442	+0.036	1.53	1.99	3.52*-2.51-1.15	3.58-0.6	2.73	2.79	1.13	1.17	4.05	3.37	0.00
81	118.00	2.50	20.0	23.0	18	68	+0.502	+0.189	1.39	1.99	3.38*-1.67-0.87	3.78+4.9	2.48	2.48	1.12	1.16	4.51	3.44	0.00
82								+0.289	1.38	1.99	3.37*-1.31-0.95		2.52	2.44	1.11	1.16	5.04	3.44	0.00
83								+0.389	1.36	1.99	3.35*-1.03-1.03		2.54	2.42	1.11	1.15	4.88	3.42	0.00
84	118.00	2.50	20.0	23.0	19	67	+0.502	+0.177	1.40	1.99	3.39*-1.56-0.83	3.53-2.0	2.57	2.57	1.16	1.21	4.82	3.52	0.00
85								+0.277	1.39	1.99	3.38*-1.24-0.91		2.61	2.53	1.16	1.20	5.37	3.53	0.00
86								+0.377	1.37	1.99	3.36*-0.99-0.99		2.63	2.51	1.15	1.20	5.23	3.50	0.00
87	118.00	2.50	20.0	23.0	19	68	-0.056	+0.089	1.48	1.99	3.47*-2.00-1.00	3.58-0.6	2.65	2.69	1.15	1.19	4.39	3.45	0.00
88								+0.189	1.46	1.99	3.45*-1.54-1.09		2.70	2.65	1.14	1.19	4.93	3.44	0.00
89								+0.289	1.44	1.99	3.43*-1.19-1.19		2.73	2.61	1.14	1.18	4.73	3.41	0.00
90	119.00	2.50	20.0	23.0	18	68	+0.937	+0.461	1.30	1.99	3.29*-0.90-0.90	3.78+4.9	2.49	2.36	1.11	1.16	5.26	3.48	0.00
91	119.00	2.50	20.0	23.0	19	67	+0.937	+0.455	1.31	1.99	3.30*-0.87-0.87	3.53-2.0	2.58	2.45	1.16	1.20	5.65	3.56	0.00
92	119.00	2.50	20.0	23.0	19	68	+0.351	+0.153	1.42	1.99	3.41*-1.65-0.87	3.58-0.6	2.59	2.59	1.16	1.21	4.72	3.51	0.00
93								+0.253	1.41	1.99	3.40*-1.31-0.95		2.63	2.56	1.15	1.20	5.27	3.51	0.00
94								+0.353	1.39	1.99	3.38*-1.04-1.04		2.65	2.53	1.15	1.19	5.11	3.49	0.00
95	119.00	2.50	20.0	23.0	19	69	-0.197	+0.071	1.50	1.99	3.49*-2.15-1.04	3.63+0.9	2.68	2.72	1.14	1.19	4.30	3.43	0.00
96								+0.171	1.48	1.99	3.47*-1.63-1.14		2.73	2.68	1.14	1.18	4.84	3.42	0.00
97								+0.271	1.45	1.99	3.44*-1.24-1.24		2.76	2.63	1.13	1.18	4.62	3.39	0.00

98	120.00	2.50	20.0	23.0	19	68	+0.779	+0.326	1.35	1.99	3.34*-1.13-0.83	3.58-0.6	2.58	2.50	1.16	1.21	5.62	3.57	0.00
99								+0.426	1.33	1.99	3.32*-0.91-0.91		2.60	2.47	1.16	1.20	5.51	3.55	0.00
100	120.00	2.50	20.0	23.0	19	69	+0.203	+0.130	1.44	1.99	3.43*-1.76-0.91	3.63+0.9	2.61	2.62	1.16	1.21	4.62	3.49	0.00
101								+0.230	1.43	1.99	3.42*-1.38-0.99		2.65	2.59	1.15	1.20	5.17	3.50	0.00
102								+0.330	1.41	1.99	3.40*-1.08-1.08		2.68	2.56	1.15	1.19	4.99	3.47	0.00
103	120.00	2.50	20.0	23.0	19	70	-0.335	+0.054	1.51	1.99	3.50*-2.31-1.09	3.68+2.3	2.71	2.76	1.14	1.19	4.20	3.41	0.00
104								+0.154	1.49	1.99	3.48*-1.72-1.19		2.76	2.71	1.13	1.18	4.74	3.40	0.00
105	120.00	2.50	20.0	23.0	20	69	-0.335	+0.031	1.52	1.99	3.51*-2.15-1.05	3.45-4.2	2.80	2.86	1.18	1.23	4.47	3.48	0.00
106								+0.131	1.50	1.99	3.49*-1.63-1.15		2.85	2.81	1.18	1.22	5.02	3.48	0.00
107	121.00	2.50	20.0	23.0	19	69	+0.623	+0.198	1.39	1.99	3.38*-1.48-0.79	3.63+0.9	2.56	2.55	1.17	1.22	4.94	3.55	0.00
108								+0.298	1.37	1.99	3.36*-1.19-0.87		2.59	2.51	1.16	1.21	5.51	3.56	0.00
109								+0.398	1.35	1.99	3.34*-0.95-0.95		2.62	2.49	1.16	1.20	5.38	3.53	0.00
110	121.00	2.50	20.0	23.0	19	70	+0.057	+0.109	1.46	1.99	3.45*-1.88-0.95	3.68+2.3	2.63	2.66	1.16	1.20	4.53	3.48	0.00
111								+0.209	1.45	1.99	3.44*-1.45-1.04		2.68	2.62	1.15	1.20	5.08	3.48	0.00
112								+0.309	1.43	1.99	3.42*-1.13-1.13		2.71	2.58	1.14	1.19	4.88	3.45	0.00
113	121.00	2.50	20.0	23.0	20	69	+0.057	+0.092	1.47	1.99	3.46*-1.76-0.91	3.45-4.2	2.72	2.75	1.20	1.25	4.82	3.55	0.00
114								+0.192	1.45	1.99	3.44*-1.38-1.00		2.77	2.71	1.19	1.24	5.38	3.56	0.00
115								+0.292	1.44	1.99	3.43*-1.09-1.09		2.80	2.68	1.19	1.23	5.20	3.53	0.00
116	121.00	2.50	20.0	23.0	20	70	-0.470	+0.015	1.54	1.99	3.53*-2.31-1.09	3.50-2.8	2.83	2.90	1.18	1.22	4.37	3.46	0.00
117	122.00	2.50	20.0	23.0	19	70	+0.470	+0.173	1.41	1.99	3.40*-1.57-0.83	3.68+2.3	2.58	2.57	1.17	1.22	4.85	3.54	0.00
118								+0.273	1.39	1.99	3.38*-1.25-0.91		2.61	2.54	1.16	1.21	5.41	3.54	0.00
119								+0.373	1.37	1.99	3.36*-0.99-0.99		2.64	2.51	1.15	1.20	5.25	3.52	0.00
120	122.00	2.50	20.0	23.0	19	71	-0.086	+0.090	1.48	1.99	3.47*-2.00-0.99	3.74+3.8	2.66	2.69	1.15	1.20	4.43	3.46	0.00
121								+0.190	1.46	1.99	3.45*-1.53-1.09		2.71	2.65	1.15	1.19	4.98	3.46	0.00
122								+0.290	1.44	1.99	3.43*-1.18-1.18		2.74	2.61	1.14	1.19	4.76	3.43	0.00
123	122.00	2.50	20.0	23.0	20	69	+0.470	+0.162	1.41	1.99	3.40*-1.47-0.80	3.45-4.2	2.67	2.66	1.21	1.26	5.15	3.61	0.00
124								+0.262	1.40	1.99	3.39*-1.19-0.87		2.70	2.62	1.20	1.25	5.74	3.62	0.00
125								+0.362	1.38	1.99	3.37*-0.95-0.95		2.72	2.59	1.20	1.24	5.61	3.60	0.00
126	122.00	2.50	20.0	23.0	20	70	-0.086	+0.071	1.49	1.99	3.48*-1.87-0.95	3.50-2.8	2.75	2.79	1.20	1.24	4.71	3.54	0.00
127								+0.171	1.47	1.99	3.46*-1.46-1.04		2.80	2.75	1.19	1.24	5.28	3.54	0.00
128								+0.271	1.45	1.99	3.44*-1.14-1.13		2.83	2.70	1.18	1.23	5.08	3.51	0.00
129	123.00	2.50	20.0	23.0	19	70	+0.903	+0.447	1.32	1.99	3.31*-0.87-0.87	3.68+2.3	2.59	2.45	1.16	1.21	5.65	3.57	0.00
130	123.00	2.50	20.0	23.0	19	71	+0.320	+0.150	1.43	1.99	3.42*-1.66-0.87	3.74+3.8	2.60	2.60	1.17	1.21	4.75	3.52	0.00
131								+0.250	1.41	1.99	3.40*-1.31-0.95		2.64	2.57	1.16	1.21	5.31	3.53	0.00

132								+0.350	1.39	1.99	3.38*-1.03-1.03		2.66	2.53	1.15	1.20	5.13	3.50	0.00
133	123.00	2.50	20.0	23.0	20	69	+0.903	+0.441	1.32	1.99	3.31*-0.84-0.84	3.45-4.2	2.67	2.53	1.20	1.25	6.04	3.65	0.00
134	123.00	2.50	20.0	23.0	20	70	+0.320	+0.136	1.44	1.99	3.43*-1.56-0.83	3.50-2.8	2.69	2.69	1.21	1.26	5.05	3.60	0.00
135								+0.236	1.42	1.99	3.41*-1.25-0.91		2.72	2.65	1.20	1.25	5.63	3.60	0.00
136								+0.336	1.40	1.99	3.39*-0.99-0.99		2.75	2.62	1.20	1.24	5.48	3.58	0.00
137	123.00	2.50	20.0	23.0	20	71	-0.226	+0.051	1.51	1.99	3.50*-2.00-0.99	3.55-1.4	2.78	2.83	1.19	1.24	4.62	3.52	0.00
138								+0.151	1.49	1.99	3.48*-1.53-1.09		2.83	2.78	1.19	1.23	5.18	3.52	0.00
139								+0.251	1.47	1.99	3.46*-1.18-1.18		2.86	2.73	1.18	1.23	4.97	3.49	0.00
140	124.00	2.50	20.0	23.0	19	71	+0.746	+0.319	1.36	1.99	3.35*-1.13-0.83	3.74+3.8	2.58	2.50	1.17	1.22	5.65	3.58	0.00
141								+0.419	1.34	1.99	3.33*-0.91-0.91		2.60	2.47	1.16	1.21	5.52	3.56	0.00
142	124.00	2.50	20.0	23.0	20	70	+0.746	+0.311	1.36	1.99	3.35*-1.08-0.80	3.50-2.8	2.67	2.58	1.21	1.26	5.99	3.65	0.00
143								+0.411	1.35	1.99	3.34*-0.87-0.87		2.69	2.55	1.20	1.25	5.89	3.64	0.00
144	124.00	2.50	20.0	23.0	20	71	+0.172	+0.113	1.46	1.99	3.45*-1.65-0.87	3.55-1.4	2.71	2.72	1.21	1.25	4.95	3.58	0.00
145								+0.213	1.44	1.99	3.43*-1.31-0.95		2.75	2.69	1.20	1.25	5.53	3.59	0.00
146								+0.313	1.42	1.99	3.41*-1.04-1.04		2.78	2.65	1.19	1.24	5.35	3.56	0.00
147	124.00	2.50	20.0	23.0	20	72	-0.363	+0.034	1.52	1.99	3.51*-2.13-1.04	3.60+0.0	2.81	2.87	1.19	1.24	4.52	3.50	0.00
148								+0.134	1.50	1.99	3.49*-1.62-1.14		2.86	2.82	1.18	1.23	5.08	3.50	0.00
149	125.00	2.50	20.0	23.0	20	71	+0.591	+0.184	1.40	1.99	3.39*-1.40-0.76	3.55-1.4	2.66	2.63	1.22	1.26	5.29	3.64	0.00
150								+0.284	1.39	1.99	3.38*-1.13-0.84		2.69	2.60	1.21	1.26	5.88	3.64	0.00
151								+0.384	1.37	1.99	3.36*-0.91-0.91		2.71	2.57	1.20	1.25	5.76	3.62	0.00
152	125.00	2.50	20.0	23.0	20	72	+0.027	+0.091	1.48	1.99	3.47*-1.76-0.91	3.60+0.0	2.73	2.76	1.20	1.25	4.85	3.57	0.00
153								+0.191	1.46	1.99	3.45*-1.38-0.99		2.77	2.72	1.20	1.25	5.43	3.57	0.00
154								+0.291	1.44	1.99	3.43*-1.08-1.08		2.80	2.68	1.19	1.24	5.23	3.55	0.00
155	125.00	2.50	20.0	23.0	21	72	-0.498	-0.006	1.55	1.99	3.54*-2.13-1.04	3.43-4.8	2.93	3.00	1.22	1.27	4.68	3.55	0.00
156	126.00	2.50	20.0	23.0	20	72	+0.439	+0.158	1.42	1.99	3.41*-1.48-0.80	3.60+0.0	2.67	2.66	1.22	1.26	5.18	3.62	0.00
157								+0.258	1.41	1.99	3.40*-1.19-0.87		2.71	2.63	1.21	1.26	5.77	3.63	0.00
158								+0.358	1.39	1.99	3.38*-0.95-0.95		2.73	2.60	1.20	1.25	5.63	3.61	0.00
159	126.00	2.50	20.0	23.0	20	73	-0.115	+0.071	1.49	1.99	3.48*-1.87-0.95	3.65+1.4	2.76	2.79	1.20	1.25	4.76	3.55	0.00
160								+0.171	1.48	1.99	3.47*-1.45-1.04		2.80	2.75	1.19	1.24	5.33	3.56	0.00
161								+0.271	1.46	1.99	3.45*-1.13-1.13		2.84	2.71	1.19	1.24	5.12	3.53	0.00
162	126.00	2.50	20.0	23.0	21	72	-0.115	+0.053	1.50	1.99	3.49*-1.75-0.91	3.43-4.8	2.85	2.88	1.24	1.29	5.04	3.62	0.00
163								+0.153	1.49	1.99	3.48*-1.38-1.00		2.89	2.84	1.24	1.28	5.63	3.63	0.00
164								+0.253	1.47	1.99	3.46*-1.09-1.09		2.92	2.80	1.23	1.28	5.44	3.60	0.00

165	127.00	2.50	20.0	23.0	20	72	+0.869	+0.433	1.33	1.99	3.32*-0.84-0.84	3.60+0.0	2.68	2.54	1.21	1.26	6.05	3.66	0.00
166	127.00	2.50	20.0	23.0	20	73	+0.289	+0.133	1.44	1.99	3.43*-1.57-0.83	3.65+1.4	2.69	2.69	1.21	1.26	5.09	3.61	0.00
167								+0.233	1.43	1.99	3.42*-1.25-0.91		2.73	2.66	1.21	1.26	5.67	3.62	0.00
168								+0.333	1.41	1.99	3.40*-0.99-0.99		2.75	2.63	1.20	1.25	5.50	3.59	0.00
169	127.00	2.50	20.0	23.0	20	74	-0.255	+0.052	1.51	1.99	3.50*-1.99-0.99	3.70+2.8	2.79	2.83	1.20	1.25	4.66	3.54	0.00
170								+0.152	1.49	1.99	3.48*-1.53-1.08		2.84	2.79	1.19	1.24	5.23	3.54	0.00
171	127.00	2.50	20.0	23.0	21	72	+0.289	+0.120	1.45	1.99	3.44*-1.48-0.80	3.43-4.8	2.78	2.77	1.26	1.30	5.39	3.68	0.00
172								+0.220	1.43	1.99	3.42*-1.19-0.88		2.81	2.74	1.25	1.30	5.99	3.69	0.00
173								+0.320	1.42	1.99	3.41*-0.96-0.96		2.84	2.71	1.24	1.29	5.85	3.67	0.00
174	127.00	2.50	20.0	23.0	21	73	-0.255	+0.032	1.52	1.99	3.51*-1.86-0.95	3.48-3.4	2.87	2.93	1.24	1.29	4.94	3.61	0.00
175								+0.132	1.50	1.99	3.49*-1.45-1.04		2.92	2.88	1.23	1.28	5.53	3.61	0.00
176								+0.232	1.48	1.99	3.47*-1.13-1.13		2.96	2.83	1.23	1.27	5.32	3.58	0.00
177	128.00	2.50	20.0	23.0	20	73	+0.713	+0.305	1.37	1.99	3.36*-1.08-0.80	3.65+1.4	2.67	2.58	1.21	1.26	6.02	3.67	0.00
178								+0.405	1.35	1.99	3.34*-0.88-0.88		2.69	2.55	1.21	1.26	5.91	3.65	0.00
179	128.00	2.50	20.0	23.0	20	74	+0.142	+0.111	1.46	1.99	3.45*-1.66-0.87	3.70+2.8	2.72	2.72	1.21	1.26	4.99	3.60	0.00
180								+0.211	1.45	1.99	3.44*-1.31-0.95		2.75	2.69	1.20	1.25	5.57	3.60	0.00
181								+0.311	1.43	1.99	3.42*-1.03-1.03		2.78	2.66	1.20	1.25	5.38	3.58	0.00
182	128.00	2.50	20.0	23.0	20	75	-0.392	+0.036	1.53	1.99	3.52*-2.12-1.03	3.75+4.2	2.82	2.87	1.19	1.24	4.56	3.52	0.00
183	128.00	2.50	20.0	23.0	21	72	+0.713	+0.298	1.38	1.99	3.37*-1.04-0.77	3.43-4.8	2.76	2.66	1.26	1.31	6.36	3.73	0.00
184								+0.398	1.36	1.99	3.35*-0.84-0.84		2.77	2.63	1.25	1.30	6.28	3.72	0.00
185	128.00	2.50	20.0	23.0	21	73	+0.142	+0.096	1.47	1.99	3.46*-1.56-0.83	3.48-3.4	2.80	2.81	1.25	1.30	5.28	3.66	0.00
186								+0.196	1.45	1.99	3.44*-1.25-0.91		2.84	2.78	1.25	1.30	5.89	3.67	0.00
187								+0.296	1.44	1.99	3.43*-1.00-1.00		2.87	2.74	1.24	1.29	5.72	3.65	0.00
188	128.00	2.50	20.0	23.0	21	74	-0.392	+0.014	1.54	1.99	3.53*-1.98-0.99	3.52-2.1	2.91	2.97	1.23	1.28	4.83	3.59	0.00
189	129.00	2.50	20.0	23.0	20	74	+0.559	+0.179	1.41	1.99	3.40*-1.41-0.76	3.70+2.8	2.66	2.64	1.22	1.27	5.31	3.65	0.00
190								+0.279	1.39	1.99	3.38*-1.14-0.84		2.69	2.61	1.21	1.26	5.92	3.66	0.00
191								+0.379	1.38	1.99	3.37*-0.91-0.91		2.71	2.58	1.21	1.26	5.77	3.64	0.00
192	129.00	2.50	20.0	23.0	20	75	-0.002	+0.090	1.48	1.99	3.47*-1.76-0.91	3.75+4.2	2.74	2.76	1.21	1.26	4.89	3.58	0.00
193								+0.190	1.46	1.99	3.45*-1.38-0.99		2.78	2.72	1.20	1.25	5.48	3.59	0.00
194								+0.290	1.44	1.99	3.43*-1.08-1.08		2.81	2.69	1.19	1.24	5.27	3.56	0.00
195	129.00	2.50	20.0	23.0	21	73	+0.559	+0.169	1.41	1.99	3.40*-1.33-0.73	3.48-3.4	2.75	2.72	1.26	1.31	5.63	3.72	0.00
196								+0.269	1.40	1.99	3.39*-1.09-0.81		2.77	2.69	1.26	1.31	6.25	3.72	0.00
197								+0.369	1.38	1.99	3.37*-0.88-0.88		2.79	2.66	1.25	1.30	6.14	3.71	0.00
198	129.00	2.50	20.0	23.0	21	74	-0.002	+0.074	1.49	1.99	3.48*-1.65-0.87	3.52-2.1	2.83	2.85	1.25	1.30	5.18	3.65	0.00
199								+0.174	1.47	1.99	3.46*-1.31-0.95		2.87	2.81	1.24	1.29	5.78	3.66	0.00

200								+0.274	1.45	1.99	3.44*-1.04-1.04		2.90	2.77	1.24	1.29	5.60	3.63	0.00
201	130.00	2.50	20.0	23.0	20	75	+0.407	+0.154	1.43	1.99	3.42*-1.49-0.80	3.75+4.2	2.68	2.66	1.22	1.27	5.21	3.64	0.00
202								+0.254	1.41	1.99	3.40*-1.19-0.87		2.71	2.63	1.21	1.26	5.81	3.64	0.00
203								+0.354	1.40	1.99	3.39*-0.95-0.95		2.74	2.60	1.21	1.26	5.65	3.62	0.00
204	130.00	2.50	20.0	23.0	21	74	+0.407	+0.142	1.44	1.99	3.43*-1.40-0.77	3.52-2.1	2.76	2.75	1.26	1.31	5.53	3.71	0.00
205								+0.242	1.42	1.99	3.41*-1.14-0.84		2.80	2.71	1.26	1.30	6.14	3.71	0.00
206								+0.342	1.40	1.99	3.39*-0.92-0.92		2.82	2.68	1.25	1.30	6.01	3.69	0.00
207	130.00	2.50	20.0	23.0	21	75	-0.144	+0.053	1.51	1.99	3.50*-1.75-0.91	3.57-0.8	2.85	2.89	1.25	1.30	5.08	3.64	0.00
208								+0.153	1.49	1.99	3.48*-1.37-0.99		2.90	2.85	1.24	1.29	5.68	3.64	0.00
209								+0.253	1.47	1.99	3.46*-1.08-1.08		2.93	2.81	1.23	1.28	5.48	3.62	0.00
210	131.00	2.50	20.0	23.0	20	75	+0.836	+0.426	1.34	1.99	3.33*-0.84-0.84	3.75+4.2	2.68	2.54	1.21	1.26	6.05	3.67	0.00
211	131.00	2.50	20.0	23.0	21	74	+0.836	+0.420	1.35	1.99	3.34*-0.81-0.81	3.52-2.1	2.76	2.62	1.26	1.31	6.44	3.74	0.00
212	131.00	2.50	20.0	23.0	21	75	+0.259	+0.118	1.46	1.99	3.45*-1.48-0.80	3.57-0.8	2.79	2.78	1.26	1.31	5.42	3.69	0.00
213								+0.218	1.44	1.99	3.43*-1.19-0.88		2.82	2.75	1.25	1.30	6.04	3.70	0.00
214								+0.318	1.42	1.99	3.41*-0.95-0.95		2.84	2.71	1.25	1.30	5.88	3.68	0.00
215	131.00	2.50	20.0	23.0	21	76	-0.283	+0.034	1.52	1.99	3.51*-1.86-0.94	3.62+0.5	2.88	2.93	1.24	1.29	4.98	3.62	0.00
216								+0.134	1.50	1.99	3.49*-1.44-1.03		2.93	2.89	1.24	1.29	5.58	3.62	0.00
217	132.00	2.50	20.0	23.0	21	75	+0.680	+0.191	1.40	1.99	3.39*-1.27-0.70	3.57-0.8	2.74	2.70	1.27	1.32	5.77	3.74	0.00
218								+0.291	1.39	1.99	3.38*-1.04-0.77		2.76	2.67	1.26	1.31	6.40	3.75	0.00
219								+0.391	1.37	1.99	3.36*-0.85-0.84		2.78	2.64	1.26	1.31	6.30	3.73	0.00
220	132.00	2.50	20.0	23.0	21	76	+0.112	+0.094	1.48	1.99	3.47*-1.56-0.83	3.62+0.5	2.81	2.82	1.26	1.31	5.32	3.68	0.00
221								+0.194	1.46	1.99	3.45*-1.25-0.91		2.85	2.78	1.25	1.30	5.93	3.69	0.00
222								+0.294	1.44	1.99	3.43*-0.99-0.99		2.87	2.75	1.24	1.29	5.75	3.66	0.00
223	132.00	2.50	20.0	23.0	21	77	-0.420	+0.016	1.54	1.99	3.53*-1.97-0.98	3.67+1.9	2.91	2.98	1.24	1.29	4.88	3.61	0.00
224	132.00	2.50	20.0	23.0	22	76	-0.420	-0.005	1.55	1.99	3.54*-1.85-0.95	3.45-4.0	3.00	3.07	1.28	1.33	5.16	3.67	0.00
225	133.00	2.50	20.0	23.0	21	76	+0.527	+0.164	1.42	1.99	3.41*-1.34-0.74	3.62+0.5	2.75	2.72	1.27	1.32	5.66	3.73	0.00
226								+0.264	1.41	1.99	3.40*-1.09-0.81		2.78	2.69	1.26	1.31	6.29	3.74	0.00
227								+0.364	1.39	1.99	3.38*-0.88-0.88		2.80	2.66	1.25	1.30	6.16	3.72	0.00
228	133.00	2.50	20.0	23.0	21	77	-0.032	+0.073	1.49	1.99	3.48*-1.65-0.87	3.67+1.9	2.83	2.86	1.26	1.31	5.22	3.67	0.00
229								+0.173	1.48	1.99	3.47*-1.31-0.95		2.87	2.82	1.25	1.30	5.83	3.67	0.00
230								+0.273	1.46	1.99	3.45*-1.03-1.03		2.90	2.78	1.24	1.29	5.63	3.65	0.00
231	133.00	2.50	20.0	23.0	22	76	-0.032	+0.057	1.50	1.99	3.49*-1.56-0.83	3.45-4.0	2.92	2.94	1.30	1.35	5.51	3.73	0.00
232								+0.157	1.48	1.99	3.47*-1.25-0.91		2.96	2.90	1.29	1.34	6.14	3.73	0.00

233								+0.257	1.47	1.99	3.46*-1.00-1.00		2.99	2.86	1.28	1.33	5.97	3.71	0.00
234	134.00	2.50	20.0	23.0	21	77	+0.376	+0.139	1.44	1.99	3.43*-1.41-0.77	3.67+1.9	2.77	2.75	1.27	1.32	5.56	3.72	0.00
235								+0.239	1.43	1.99	3.42*-1.14-0.84		2.80	2.72	1.26	1.31	6.18	3.72	0.00
236								+0.339	1.41	1.99	3.40*-0.91-0.91		2.82	2.69	1.25	1.30	6.03	3.71	0.00
237	134.00	2.50	20.0	23.0	21	78	-0.173	+0.053	1.51	1.99	3.50*-1.75-0.90	3.71+3.2	2.86	2.90	1.25	1.30	5.13	3.65	0.00
238								+0.153	1.49	1.99	3.48*-1.37-0.99		2.91	2.86	1.25	1.30	5.74	3.66	0.00
239								+0.253	1.47	1.99	3.46*-1.07-1.07		2.94	2.81	1.24	1.29	5.52	3.63	0.00
240	134.00	2.50	20.0	23.0	22	76	+0.376	+0.128	1.45	1.99	3.44*-1.33-0.74	3.45-4.0	2.85	2.83	1.31	1.36	5.87	3.78	0.00
241								+0.228	1.43	1.99	3.42*-1.09-0.81		2.88	2.80	1.30	1.35	6.51	3.79	0.00
242								+0.328	1.42	1.99	3.41*-0.88-0.88		2.90	2.77	1.29	1.34	6.38	3.77	0.00
243	134.00	2.50	20.0	23.0	22	77	-0.173	+0.035	1.52	1.99	3.51*-1.65-0.87	3.50-2.8	2.94	2.98	1.29	1.34	5.41	3.71	0.00
244								+0.135	1.50	1.99	3.49*-1.31-0.95		2.99	2.94	1.29	1.34	6.04	3.72	0.00
245								+0.235	1.48	1.99	3.47*-1.03-1.03		3.02	2.90	1.28	1.33	5.84	3.70	0.00
246	135.00	2.50	20.0	23.0	21	77	+0.802	+0.313	1.37	1.99	3.36*-1.00-0.75	3.67+1.9	2.75	2.65	1.27	1.32	6.55	3.77	0.00
247								+0.413	1.36	1.99	3.35*-0.81-0.81		2.77	2.62	1.26	1.31	6.45	3.75	0.00
248	135.00	2.50	20.0	23.0	21	78	+0.228	+0.115	1.46	1.99	3.45*-1.48-0.80	3.71+3.2	2.79	2.78	1.26	1.32	5.46	3.71	0.00
249								+0.215	1.45	1.99	3.44*-1.19-0.87		2.83	2.75	1.26	1.31	6.08	3.71	0.00
250								+0.315	1.43	1.99	3.42*-0.95-0.95		2.85	2.72	1.25	1.30	5.90	3.69	0.00
251	135.00	2.50	20.0	23.0	21	79	-0.312	+0.035	1.53	1.99	3.52*-1.85-0.94	3.76+4.5	2.89	2.94	1.25	1.30	5.03	3.64	0.00
252								+0.135	1.51	1.99	3.50*-1.43-1.02		2.94	2.89	1.24	1.29	5.64	3.64	0.00
253	135.00	2.50	20.0	23.0	22	76	+0.802	+0.307	1.38	1.99	3.37*-0.96-0.72	3.45-4.0	2.83	2.72	1.31	1.36	6.90	3.83	0.00
254								+0.407	1.36	1.99	3.35*-0.78-0.78		2.84	2.69	1.30	1.35	6.83	3.82	0.00
255	135.00	2.50	20.0	23.0	22	77	+0.228	+0.102	1.47	1.99	3.46*-1.40-0.77	3.50-2.8	2.87	2.86	1.31	1.36	5.76	3.77	0.00
256								+0.202	1.45	1.99	3.44*-1.14-0.84		2.91	2.83	1.30	1.35	6.40	3.77	0.00
257								+0.302	1.44	1.99	3.43*-0.92-0.92		2.93	2.80	1.29	1.34	6.25	3.76	0.00
258	135.00	2.50	20.0	23.0	22	78	-0.312	+0.015	1.53	1.99	3.52*-1.74-0.90	3.55-1.5	2.98	3.03	1.29	1.34	5.31	3.70	0.00
259								+0.115	1.52	1.99	3.51*-1.37-0.99		3.02	2.98	1.28	1.33	5.93	3.71	0.00
260	136.00	2.50	20.0	23.0	21	78	+0.648	+0.185	1.41	1.99	3.40*-1.28-0.71	3.71+3.2	2.74	2.70	1.27	1.33	5.79	3.75	0.00
261								+0.285	1.39	1.99	3.38*-1.04-0.78		2.77	2.67	1.27	1.32	6.44	3.76	0.00
262								+0.385	1.38	1.99	3.37*-0.85-0.85		2.78	2.64	1.26	1.31	6.31	3.74	0.00
263	136.00	2.50	20.0	23.0	21	79	+0.082	+0.093	1.48	1.99	3.47*-1.56-0.83	3.76+4.5	2.82	2.82	1.26	1.31	5.36	3.69	0.00
264								+0.193	1.46	1.99	3.45*-1.25-0.91		2.85	2.79	1.26	1.31	5.98	3.70	0.00
265								+0.293	1.44	1.99	3.43*-0.99-0.99		2.88	2.75	1.25	1.30	5.78	3.68	0.00
266	136.00	2.50	20.0	23.0	22	77	+0.648	+0.178	1.41	1.99	3.40*-1.21-0.68	3.50-2.8	2.82	2.78	1.31	1.37	6.12	3.82	0.00

267								+0.278	1.40	1.99	3.39*-1.00-0.75		2.84	2.74	1.31	1.36	6.78	3.82	0.00
268								+0.378	1.38	1.99	3.37*-0.82-0.82		2.86	2.71	1.30	1.35	6.69	3.81	0.00
269	136.00	2.50	20.0	23.0	22	78	+0.082	+0.078	1.49	1.99	3.48*-1.48-0.80 3.55-1.5		2.90	2.90	1.30	1.36	5.66	3.75	0.00
270								+0.178	1.47	1.99	3.46*-1.19-0.88		2.93	2.87	1.30	1.35	6.30	3.76	0.00
271								+0.278	1.45	1.99	3.44*-0.95-0.95		2.96	2.83	1.29	1.34	6.13	3.74	0.00
272	136.00	2.50	20.0	23.0	22	79	-0.448	-0.003	1.55	1.99	3.54*-1.84-0.94 3.59-0.3		3.01	3.08	1.29	1.34	5.21	3.68	0.00
273	137.00	2.50	20.0	23.0	21	79	+0.495	+0.159	1.43	1.99	3.42*-1.34-0.74 3.76+4.5		2.76	2.73	1.27	1.32	5.69	3.74	0.00
274								+0.259	1.41	1.99	3.40*-1.09-0.81		2.79	2.69	1.26	1.32	6.33	3.75	0.00
275								+0.359	1.40	1.99	3.39*-0.88-0.88		2.81	2.66	1.26	1.31	6.18	3.73	0.00
276	137.00	2.50	20.0	23.0	22	78	+0.495	+0.150	1.43	1.99	3.42*-1.27-0.71 3.55-1.5		2.84	2.80	1.31	1.37	6.01	3.80	0.00
277								+0.250	1.42	1.99	3.41*-1.04-0.78		2.87	2.77	1.31	1.36	6.66	3.81	0.00
278								+0.350	1.40	1.99	3.39*-0.85-0.85		2.88	2.74	1.30	1.35	6.54	3.79	0.00
279	137.00	2.50	20.0	23.0	22	79	-0.061	+0.056	1.50	1.99	3.49*-1.56-0.83 3.59-0.3		2.92	2.95	1.30	1.35	5.56	3.74	0.00
280								+0.156	1.49	1.99	3.48*-1.24-0.91		2.96	2.91	1.29	1.35	6.19	3.75	0.00
281								+0.256	1.47	1.99	3.46*-0.99-0.99		2.99	2.87	1.29	1.34	6.00	3.73	0.00
282	138.00	2.50	20.0	23.0	21	79	+0.926	+0.434	1.34	1.99	3.33*-0.78-0.78 3.76+4.5		2.75	2.60	1.26	1.32	6.60	3.78	0.00
283	138.00	2.50	20.0	23.0	22	78	+0.926	+0.430	1.35	1.99	3.34*-0.76-0.76 3.55-1.5		2.83	2.68	1.31	1.36	7.00	3.84	0.00
284	138.00	2.50	20.0	23.0	22	79	+0.345	+0.124	1.45	1.99	3.44*-1.34-0.74 3.59-0.3		2.86	2.83	1.31	1.37	5.90	3.79	0.00
285								+0.224	1.44	1.99	3.43*-1.09-0.81		2.89	2.80	1.31	1.36	6.55	3.80	0.00
286								+0.324	1.42	1.99	3.41*-0.88-0.88		2.91	2.77	1.30	1.35	6.41	3.78	0.00
287	138.00	2.50	20.0	23.0	22	80	-0.202	+0.035	1.52	1.99	3.51*-1.64-0.86 3.64+1.0		2.95	2.99	1.30	1.35	5.46	3.73	0.00
288								+0.135	1.50	1.99	3.49*-1.30-0.95		3.00	2.95	1.29	1.34	6.09	3.73	0.00
289								+0.235	1.48	1.99	3.47*-1.03-1.03		3.03	2.91	1.28	1.34	5.88	3.71	0.00
290	138.00	2.50	20.0	23.0	23	79	-0.202	+0.018	1.53	1.99	3.52*-1.56-0.83 3.43-4.6		3.03	3.07	1.34	1.39	5.74	3.78	0.00
291								+0.118	1.51	1.99	3.50*-1.25-0.91		3.08	3.03	1.33	1.38	6.39	3.79	0.00
292								+0.218	1.49	1.99	3.48*-1.00-0.99		3.11	2.99	1.32	1.38	6.21	3.77	0.00
293	139.00	2.50	20.0	23.0	22	79	+0.769	+0.300	1.38	1.99	3.37*-0.96-0.72 3.59-0.3		2.83	2.73	1.31	1.36	6.93	3.84	0.00
294								+0.400	1.37	1.99	3.36*-0.79-0.79		2.85	2.70	1.31	1.36	6.85	3.83	0.00
295	139.00	2.50	20.0	23.0	22	80	+0.198	+0.099	1.47	1.99	3.46*-1.41-0.77 3.64+1.0		2.88	2.87	1.31	1.36	5.80	3.78	0.00
296								+0.199	1.46	1.99	3.45*-1.14-0.84		2.91	2.84	1.30	1.36	6.45	3.79	0.00
297								+0.299	1.44	1.99	3.43*-0.92-0.92		2.94	2.80	1.30	1.35	6.28	3.77	0.00
298	139.00	2.50	20.0	23.0	22	81	-0.341	+0.016	1.54	1.99	3.53*-1.73-0.90 3.68+2.3		2.98	3.04	1.29	1.35	5.36	3.71	0.00
299								+0.116	1.52	1.99	3.51*-1.36-0.98		3.03	2.99	1.29	1.34	5.99	3.72	0.00
300	139.00	2.50	20.0	23.0	23	79	+0.198	+0.087	1.48	1.99	3.47*-1.34-0.74 3.43-4.6		2.96	2.95	1.35	1.40	6.10	3.84	0.00

конец протокола lines: 760